

QMST5336: ANALYTICS

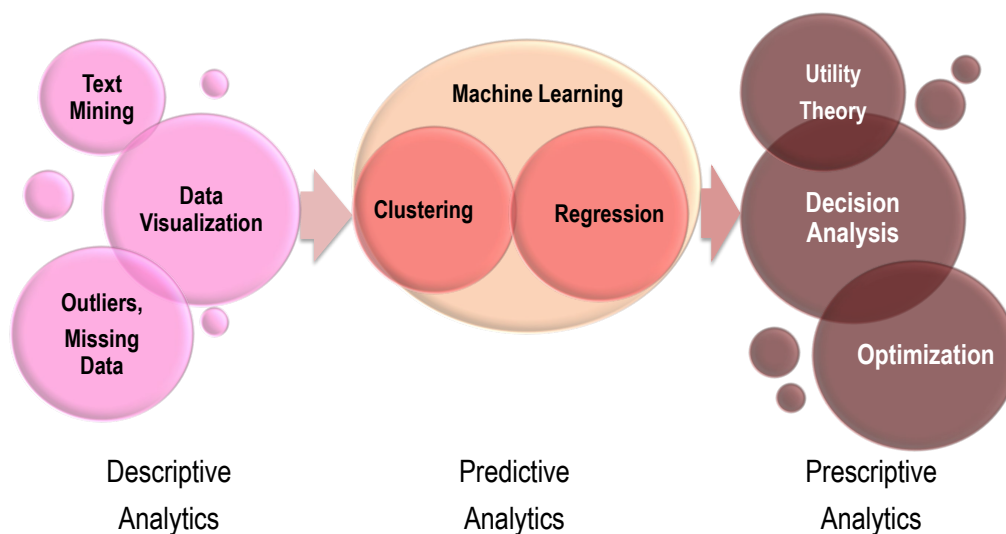
Lecture 9: Optimization

Emily Zhu, PhD.

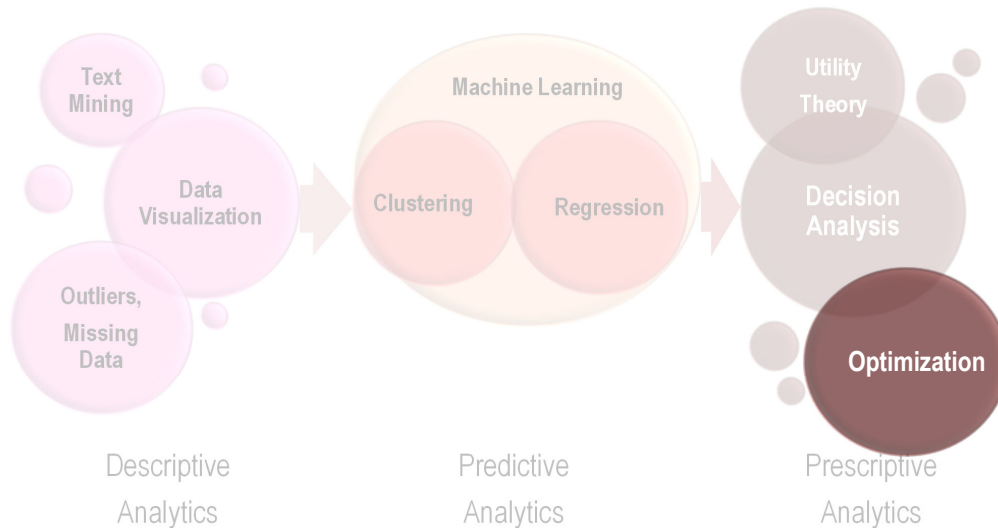
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Our Problem and Roadmap

An investment property in Austin: **Airbnb** versus **Rental**?



Roadmap: Where We Are



Case: One-off Utensils in the Airbnb Property

It is time consuming to clear and set up the house after every stay. In order to save cleaning times, we can use one-off utensils made in paper, plastic or bamboo. However, compared with ceramic/glass utensils, one-off ones are

- more costly
- not environment friendly and generating more garbage



Utensils	Labor(hr/stay)	Garbage(lb/stay)	Profit(\$/stay)
One-off	1	4	200
Ceramic	2	3	250

There are 20 hours of labor and 60 lbs of garbage allowance to spare per month.

Question 1: To maximize monthly profits, how many stays should we choose one-off utensils?

Linear Programming (LP) Model

Decision variables

- symbols representing levels or quantities of activity/actions

Objective function

- linear relationship of decision variables for the objective of a business
- most frequent business objective is to maximize profit
- most frequent objective of individual operational units (such as a production or packaging department) is to minimize cost

Constraint

- linear relationship representing a restriction on decision making

LP Model Formulation

- **Decision variables** x_j
- **Objective function** Max/Min $z = (c_1x_1 + c_2x_2 + \dots + c_nx_n)$
- **Constraints** subject to:

$$\left\{ \begin{array}{ll} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n (\leq, =, \geq) & b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n (\leq, =, \geq) & b_2 \\ & \dots \\ & \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n (\leq, =, \geq) & b_n \end{array} \right.$$

where

- b_i are constraint levels,
- c_j are objective function coefficients,
- a_{ij} are constraint coefficients.

Case: LP Model Formulation

Utensils	Labor(hr/stay)	Garbage(lb/stay)	Profit(\$/stay)
One-off	1	4	200
Ceramic	2	3	250

There are 20 hours of labor and 60 lbs of garbage allowance to spare per month.

Decision variables

x_1 : the number of stays using one-off utensils per month;

x_2 : the number of stays using ceramic utensils per month.

Objective function Maximize $Z = \$200 x_1 + 250x_2$

Constraints Subject to

$$x_1 + 2x_2 \leq 20 \text{ hr}$$

$$4x_1 + 3x_2 \leq 60 \text{ lb}$$

$$x_1, x_2 \geq 0$$

$$x_1 + x_2 \leq 30$$

Example 1: Formulate A Minimization Problem

M&D Chemicals produces two products that are sold as raw materials to companies that manufacture bath soaps and laundry detergents. Based on an analysis of current inventory levels and potential demand for the coming month, M&D's management specified that the combined production for products A and B must total at least 14,000 gallons. Separately, a major customer's order for 8,000 gallons consisting of product A and product B with ratio 1.25:1. M&D's objective is to satisfy these requirements at a minimum total production cost. Production costs are \$6 per gallon for product A and \$3 per gallon for product B.

Decision variables

x_1 : the amount of product A to produce in thousand gallons;

x_2 : the amount of product B to produce in thousand gallons;

Objective function Minimize $Z = \$ 6x_1 + 3x_2$

Constraints Subject to

$$x_1 + x_2 \geq 14$$

$$x_1 \geq 8 \frac{1.25}{1.25 + 1}$$

$$x_2 \geq 8 \frac{1}{1.25 + 1}$$

Example 2: LP Model Formulation

Round Tree Manor is a hotel that provides two types of rooms with three rental classes: Super Saver, Deluxe, and Business. The profit per night for each type of room and rental class is as follows:

	Rental Class (\$)		
Room	Super Saver	Deluxe	Business
Type I	30	35	-
Type II	20	30	40

Round Tree's management makes a forecast of the demand by rental class for each night in the future: 130 rentals in the Super Saver class, 60 in the Deluxe class, and 50 in the Business class. Round Tree has 100 Type I rooms and 120 Type II rooms. Can you develop a linear programming model to determine the amount of reservations to accept for each rental class in order to maximize profit per night?

Example 2: C'td

	Rental Class (\$)		
Room	Super Saver	Deluxe	Business
Type I	30	35	-
Type II	20	30	40

130 rentals in the Super Saver class, 60 in the Deluxe class, and 50 in the Business class. Round Tree has 100 Type I rooms and 120 Type II rooms.

Decision variables x_{1S} : the amount of rental class of room types;

Room	Super Saver	Deluxe	Business
Type I	x_{1S}	x_{1D}	-
Type II	x_{2S}	x_{2D}	x_{2B}

Objective function Maximize $Z = 30x_{1S} + 35x_{1D} + 20x_{2S} + 30x_{2D} + 40x_{2B}$

Constraints Subject to

$$x_{1S} + x_{1D} \leq 100$$

$$x_{2S} + x_{2D} + x_{2B} \leq 120$$

$$x_{1S} + x_{2S} \leq 130$$

$$x_{1D} + x_{2D} \leq 60$$

$$x_{2B} \leq 50$$

Binary Variables: Capital Budgeting

Capital budgeting problem: A binary integer programming problem that involves choosing which possible projects or activities provide the best investment return. In a capital budgeting problem, the objective function is to maximize the net present value of the capital budgeting projects.

The Ice-Cold Refrigerator Company is considering investing in several projects that have varying capital requirements over the next four years. Faced with limited capital each year, management would like to select the most profitable projects that it can afford.

Project Net Present Value, Capital Requirements, and Available Capital for the Ice-Cold Refrigerator Company

	Plant Expan (\$)	Warehouse Expan (\$)	New Machine (\$)	New Pro. Research (\$)	Total Cap. Available (\$)
Present V.	90,000	40,000	10,000	37,000	
Y1 Cap	15,000	10,000	10,000	15,000	40,000
Y2 Cap	20,000	15,000		10,000	50,000
Y3 Cap	20,000	20,000		10,000	40,000
Y4 Cap	15,000	5,000	4,000	10,000	35,000

Binary Variables: Capital Budgeting 2

Let us define four binary decision variables:

$P = 1$ if the plant expansion project is accepted; 0 if rejected.

$W = 1$ if the warehouse expansion project is accepted; 0 if rejected.

$M = 1$ if the new machinery project is accepted; 0 if rejected.

$R = 1$ if the new product research project is accepted; 0 if rejected.

This problem has four constraints:

One for the funds available in each of the next four years.

A binary integer linear programming model (thousands of dollars):

Max $90P + 40W + 10M + 37R$

s.t.

$15P + 10W + 10M + 15R \leq 40$ (Year 1 capital available)

$20P + 15W + 5R \leq 50$ (Year 2 capital available)

$20P + 20W + 10R \leq 40$ (Year 3 capital available)

$15P + 5W + 4M + 10R \leq 35$ (Year 4 capital available)

$P, W, M, R = 0, 1$

LP Solvers

Free & Pedagogical Use

pymprog

PyMathProg 1.0 documentation

[NEXT](#) | [INDEX](#)

Welcome to PyMathProg

Free

apply to nonlinear optimization



[SciPy.org](#)

[Docs](#)

[SciPy v1.3.1 Reference Guide](#)

Optimization and Root Finding
([scipy.optimize](#))

Commercial (\$\$\$)

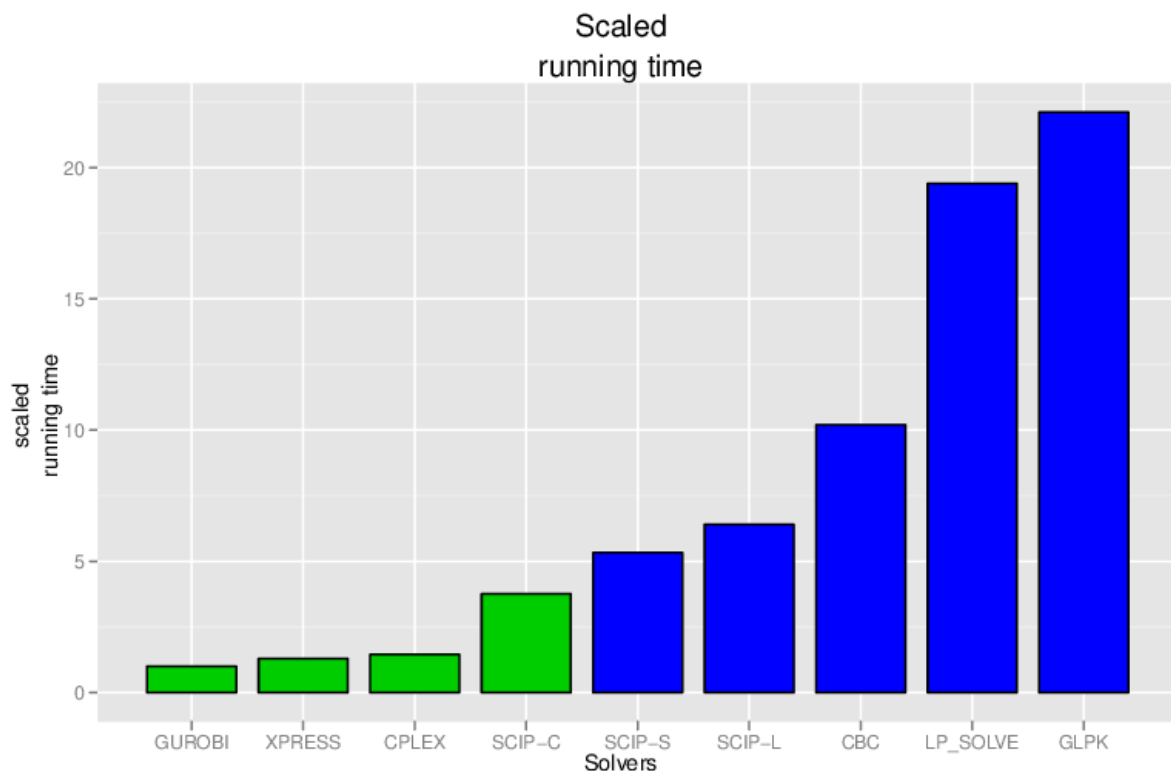


Commercial (\$\$\$)

apply to quadratic optimization



Commercial versus Non-commercial Solvers



Solve LP with Python

```
1 # import library
2 import pymprog as pm
3
4 pm.begin('Airbnb_Rental')
5 x, y = pm.var('x, y') # variables
6 pm.maximize(200 * x + 250 * y, 'profit')
7 x + 2*y <= 20 # Labor
8 4*x + 3*y <= 60 # Garbage
9 x + y <= 30 # Days in a month
10 pm.solve()
11 print('\nMax profit:', pm.vobj())
12 pm.sensitivity()
13 pm.end()
```

Max profit: 3400.0

PyMathProg 1.0 Sensitivity Report Created: 2019/11/03 Sun 19:33PM

Variable	Activity	Dual.Value	Obj.Coeff	Range.From	Range.Till
*x1	12	0	200	125	333.333
*x2	4	0	250	150	400

Constraint	Activity	Dual.Value	Lower.Bnd	Upper.Bnd	RangeLower	RangeUpper
R1	20	80	-inf	20	15	40
R2	60	30	-inf	60	30	80
*R3	16	0	-inf	30	16	16

Binary Variables in Python

```
1 pm.begin('CapitalBudgeting')
2 #pm.verbose(True)
3 P, W, M, R = pm.var('P, W, M, R', kind = bool) # variables
4 pm.maximize(90 * P + 40 * W + 10 * M + 37 * R, 'present value')
5 15 * P + 10 * W + 10 * M + 15 * R <= 40 #Year 1
6 20 * P + 15 * W + 5 * R <= 50 # Year 2
7 20 * P + 20 * W + 10 * R <= 40 # Year 3
8 15 * P + 5 * W + 4 * M + 10 * R <= 35 # Year 4
9 pm.solve()
10 print('\nMax present value:', pm.vobj())
11 print('\nP:', P.primal)
12 print('\nW:', W.primal)
13 print('\nM:', M.primal)
14 print('\nR:', R.primal)
15 pm.end()
```

Max present value: 140.0

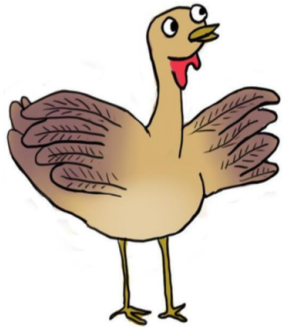
P: 1.0

W: 1.0

M: 1.0

R: 0.0

Why Sensitivity Report?



Tom

Why do we need to use a report? Can't we just solve the problem many times using Solver?



Cathy

Solving the problem multiple times is OK for small spreadsheets, but there are advantages of understanding the SR.

The SR is a compact way of storing information.

LP Solvers generate lots of useful information with a single report.

Solving multiple times is not practical if there are 1000s of constraints.

In addition, it is needed to understand LP theory.

Finally, it will be on Quiz 3.

Emily Zhu, PhD.

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Sensitivity Report

The structure of the optimal solution stays the same with small changes of data.

Max/Min $z = (c_1x_1 + c_2x_2 + \dots + c_nx_n)$
subject to:

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n (\leq, =, \geq) & b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n (\leq, =, \geq) & b_2 \\ \dots & \dots \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n (\leq, =, \geq) & b_n \end{cases}$$

Changes that we consider

- 1 change the objective coefficient c_j of a variable **Range.From/Till**
- 2 change the constraint levels b_i of a constraint
 - changing the initial conditions **RangeLower/Upper**
 - purchasing resources at a cost **Dual. Value (= Shadow Price)**

Sensitivity Report C'td

Changes that we consider

- 1 change the objective coefficient c_j of a variable **Range.From/Till**
- 2 change the constraint levels b_i of a constraint
 - changing the initial conditions **RangeLower/Upper**
 - purchasing resources at a cost **Dual. Value (= Shadow Price)**

PyMathProg 1.0 Sensitivity Report Created: 2019/11/03 Sun 19:33PM

Variable	Activity	Dual.Value	Obj.Coef	Range.From	Range.Till
*x1	12	0	200	125	333.333
*x2	4	0	250	150	400

Constraint	Activity	Dual.Value	Lower.Bnd	Upper.Bnd	RangeLower	RangeUpper
R1	20	80	-inf	20	15	40
R2	60	30	-inf	60	30	80
*R3	16	0	-inf	30	16	16

Shadow Prices

Shadow Price

The **shadow price** of a constraint of a linear program is the increase in the optimal objective value per unit increase in the constraint level, i.e. b_i .

Shadow Price is valid

only if the constraint level satisfies allowable ranges, i.e. *RangeLower/Upper*.

The Sensitivity Report does NOT say

- what the new solution is;
- what happens if the change is beyond the allowable range.

Non-binding

If an inequality constraint does not hold with equality in the optimum solution, the constraint is **non-binding**.

The shadow price of each non-binding constraint is 0.

Shadow Prices: Example

PyMathProg 1.0 Sensitivity Report Created: 2019/11/03 Sun 19:33PM

Variable	Activity	Dual.Value	Obj.Coef	Range.From	Range.Till
*x1	12	0	200	125	333.333
*x2	4	0	250	150	400

Constraint	Activity	Dual.Value	Lower.Bnd	Upper.Bnd	RangeLower	RangeUpper
R1	20	80	-inf	20	15	40
R2	60	30	-inf	60	30	80
*R3	16	0	-inf	30	16	16

For example the shadow price of labor is 80\$ per hour, which is valid if the monthly labor to spare is between 15 and 40 hours.

the shadow price of garbage is 30\$/lb, which is valid if the monthly garbage allowance is between 30 and 80 lbs.

the shadow price of day of month is 0 and this is non-binding constraint.

Case: Nonlinear Programming Model

$$Demand = 46 - \frac{Daily\ Price\ (\$)}{10} \quad (days)$$

Utensils	Labor(hr/stay)	Garbage(lb/stay)	Cost(\$/stay)
One-off	1	4	50
Ceramic	2	3	40

There are 20 hours of labor and 60 lbs of garbage allowance to spare per month.

Question 2: To maximize monthly profits, how much should we charge per night? And how many stays should we choose one-off utensils?

Decision variables

x_1 : the number of stays using one-off utensils per month;

x_2 : the number of stays using ceramic utensils per month;

x_3 : daily price (\$).

Objective function Maximize $Z = x_1(x_3 - 50) + x_2(x_3 - 40)$

Case: Formulate Nonlinear Programming Model

Utensils	Labor(hr/stay)	Garbage(lb/stay)	Cost(\$/stay)
One-off	1	4	50
Ceramic	2	3	40

There are 20 hours of labor and 60 lbs of garbage allowance to spare per month.

Decision variables

x_1 : the number of stays using one-off utensils per month;

x_2 : the number of stays using ceramic utensils per month;

x_3 : daily price (\$).

Constraints Subject to

$$x_1 + 2x_2 \leq 20 \text{ hr}$$

$$4x_1 + 3x_2 \leq 60 \text{ lb}$$

$$x_1, x_2, x_3 \geq 0$$

$$x_1 + x_2 = 46 - x_3/10 \leq 30$$

Standardize Formulation

Minimize $z = (c_1x_1 + c_2x_2 + \dots + c_nx_n)$

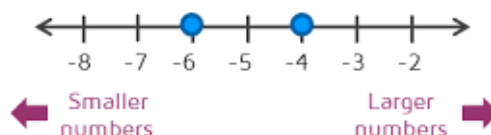
subject to: **Constraints**

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n - b_1(=, \geq) & 0 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n - b_2(=, \geq) & 0 \\ \dots & \\ \dots & \\ a_{n1}x_1 + a_{n2}x_2 + \dots + a_{nn}x_n - b_n(=, \geq) & 0 \end{cases}$$

What about "Max" objective functions?

$$-1 \cdot 4 < 6 \cdot -1$$

$$-4 \not< -6$$



Case: Reformulate the Nonlinear Program Model

Decision variables

x_1 : the number of stays using one-off utensils per month;

x_2 : the number of stays using ceramic utensils per month;

x_3 : daily price (\$).

Maximize $x_1(x_3 - 50) + x_2(x_3 - 40)$

Subject to

$$x_1 + 2x_2 \leq 20$$

$$4x_1 + 3x_2 \leq 60$$

$$x_1, x_2, x_3 \geq 0$$

$$x_1 + x_2 = 46 - x_3/10 \leq 30$$

Reformulate

Minimize $-(x_1(x_3 - 50) + x_2(x_3 - 40))$

Subject to $20 - (x_1 + 2x_2) \geq 0$

$$60 - (4x_1 + 3x_2) \geq 0$$

$$46 - x_3/10 - (x_1 + x_2) = 0$$

$$160 \leq x_3 \leq 460$$

$$0 \leq 46 - \frac{x_3}{10} \leq 30 \Rightarrow 160 \leq x_3 \leq 460$$

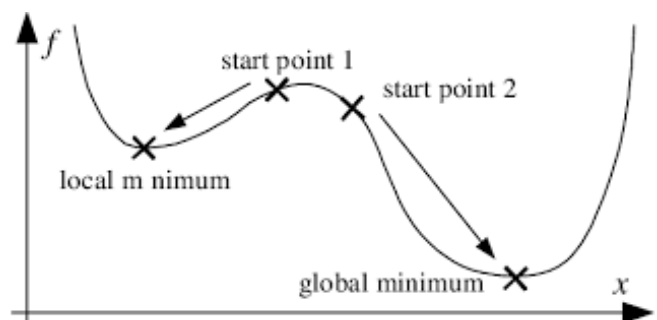
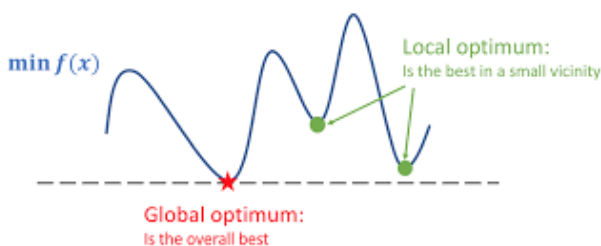
NLP Code

```
1 import numpy as np
2 from scipy.optimize import minimize
3
4 def objective(x):
5     return -((x[2] - 50)*x[0] + (x[2] - 40)*x[1])
6 def constraint1(x):
7     return 20 - (x[0] + 2*x[1])
8 def constraint2(x):
9     return 60 - (x[0]*4 + x[1]*3)
10 def constraint3(x):
11     return x[0] + x[1] + x[2]/10 - 46
12 # initial guesses
13 n = 3
14 x0 = np.zeros(n)
15 x0[2] = 200
16 # optimize
17 b = (0.0, 30.0)
18 bnds = (b, b, (160, 460))
19 con1 = {'type': 'ineq', 'fun': constraint1}
20 con2 = {'type': 'ineq', 'fun': constraint2}
21 con3 = {'type': 'eq', 'fun': constraint3}
22 cons = ([con1, con2, con3])
23 solution = minimize(objective, x0, bounds = bnds, constraints=cons)
24 print(solution)
```

```
Initial Objective: -0.0
  fun: -4040.000000005808
  jac: array([-250., -260., -16.])
message: 'Optimization terminated successfully.'
  nfev: 15
   nit: 3
  njev: 3
status: 0
success: True
   x: array([ 12.,   4., 300.])
Final Objective: -4040.000000005808
Solution
x1 = 12.000000000000357
x2 = 3.9999999999995453
```

Local versus Global Optimums: When A Solver Fails

```
Initial Objective: -0.0
  fun: -3400.00000000384
  jac: array([-200., -250.])
message: 'Positive directional derivative for linesearch'
  nfev: 26
   nit: 8
  njev: 4
status: 8
success: False
   x: array([12.,  4.])
Final Objective: -3400.00000000384
Solution
x1 = 11.99999999998274
x2 = 4.000000000024739
```



Solution: Try different initial guesses

Example: Nonlinear Optimization 1/6

Harry Markowitz received the 1990 Nobel Prize for his ground-breaking work in portfolio optimization. Money management firms throughout the world use numerous variations of this basic Markowitz mean-variance portfolio model. A key trade-off in financial planning is that between risk and return. For a chance to earn greater returns, the investor must also accept greater risk. In most portfolio optimization models, the return used is the expected (or average) return of the possible outcomes, and the risk is some measure of variability in these possible outcomes.

It is impossible to predict the exact return for any stocks over the next 12 months. The following table shows the four different scenarios of the annual return (%) for Apple Computer (APPL), Advanced Micro Devices (AMD) and Oracle Corporation (ORCL). Assume that the four scenarios are equally likely to happen.

Scenario	1	2	3	4
APPL	-39.8	10.10	124.90	151.80
AMD	-42.50	13.60	56.90	36.70
ORCL	-10.20	137.90	170.60	16.60

Example: Measure Risk Using Variance 2/6

For the purpose of building portfolios for their clients, portfolio managers have to determine the proportion of the portfolio that will be invested in each of the stocks. We use the following decision variables:

- AL = the proportion of the portfolio invested in stock APPL
- AD = the proportion of the portfolio invested in stock AMD
- OL = the proportion of the portfolio invested in stock ORCL

Because the sum of these proportions must equal one, we need the following constraint:

$$AL + AD + OL = 1$$

The portfolio return over the next 12 months depends on which of the possible scenarios in the above table occurs. Let R_1 denote the portfolio return if the scenario 1 occurs, R_2 denote the portfolio return if the scenario 2 occurs, and so on.

- Scenario 1 return: $R_1 = -39.80AL - 42.50AD - 10.20OL$
- Scenario 2 return: $R_2 = 10.10AL + 13.60AD + 137.90OL$
- Scenario 3 return: $R_3 = 124.90AL + 56.90AD + 170.60OL$
- Scenario 4 return: $R_4 = 151.80AL + 36.70AD + 16.60OL$

Example: Measure Risk Using Variance 3/6

Because the four planning scenarios in the model are equally likely to occur, then the expected return for the portfolio $\bar{R}$ is

$$\bar{R} = \frac{1}{4}(R_1 + R_2 + R_3 + R_4)$$

. The measure of risk most often associated with the Markowitz portfolio model is the variance of the portfolio's return

$$Var = \frac{1}{4} \sum_{i=1}^4 (R_i - \bar{R})^2.$$

That is, the portfolio variance is the average of the sum of the squares of the deviations from the mean value under each scenario. The larger this number, the more widely dispersed the scenario returns are about the average value. If the portfolio variance were equal to zero, then every scenario return R_i would be equal, and there would be no risk.

Please develop nonlinear programming models to address the following question:

- What should portfolio managers do to minimize the risk of a portfolio with a required expected return of 35%?

Example: Measure Risk Using Variance 4/6

Decision variables:

- AL = the proportion of the portfolio invested in stock APPL
- AD = the proportion of the portfolio invested in stock AMD
- OL = the proportion of the portfolio invested in stock ORCL

minimize the risk of a portfolio with a required expected return

Objective:

$$\min \quad \frac{1}{4} \sum_{i=1}^4 (R_i - \bar{R})^2$$

Constraints:

$$R_1 = -39.80AL - 42.50AD - 10.20OL \text{ Scenario 1}$$

$$R_2 = 10.10AL + 13.60AD + 137.90OL \text{ Scenario 2}$$

$$R_3 = 124.90AL + 56.90AD + 170.60OL \text{ Scenario 3}$$

$$R_4 = 151.80AL + 36.70AD + 16.60OL \text{ Scenario 4}$$

$$\bar{R} = \frac{1}{4}(R_1 + R_2 + R_3 + R_4) \text{ define expected return}$$

$$AL + AD + OL = 1 \text{ sum of proportions is one}$$

$$\bar{R} \geq 35 \text{ a required expected return}$$

When coding for the solver, we use the following notations $x_0 : AL$, $x_1 : AD$, $x_2 : OL$, $x_3 : R_1$, $x_4 : R_2$, $x_5 : R_3$, $x_6 : R_4$, $x_7 : \bar{R}$.

Code: Measure Risk Using Variance 5/6

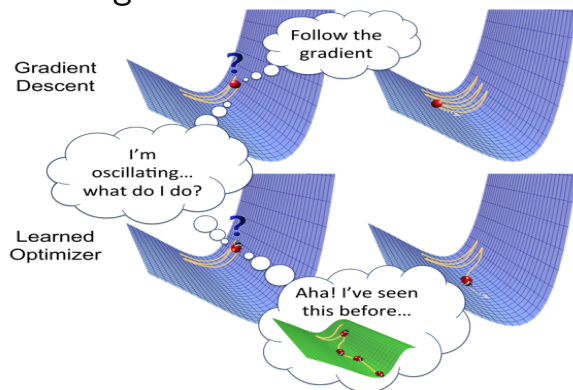
```
1 def objective(x):
2     return 1/4*((x[3] - x[7])**2 + (x[4] - x[7])**2 + (x[5] - x[7])
3         **2 + (x[6] - x[7])**2)
4
5 def constraint1(x):
6     return -39.8*x[0] - 42.5*x[1] - 10.2*x[2] - x[3]
7
8 def constraint2(x):
9     return 10.1*x[0] + 13.6*x[1] + 137.9*x[2] - x[4]
10
11 def constraint3(x):
12     return 124.9*x[0] + 56.9*x[1] + 170.6*x[2] - x[5]
13
14 def constraint4(x):
15     return 151.8*x[0] + 36.7*x[1] + 16.6*x[2] - x[6]
16
17 def constraint5(x):
18     return x[0] + x[1] + x[2] - 1
19
20 def constraint6(x):
21     return (x[3] + x[4] + x[5] + x[6])/4 - x[7]
```

Code: Measure Risk Using Variance 6/6

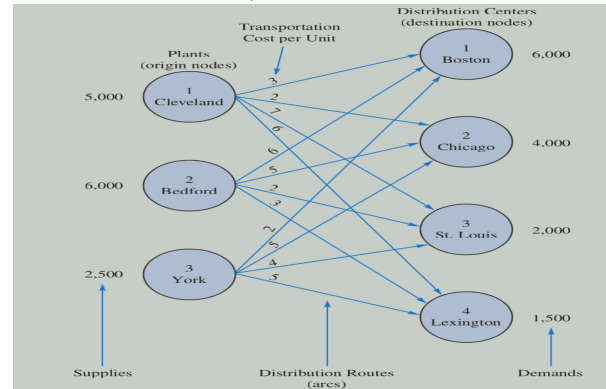
```
1 # initial guesses
2 n = 8
3 x0 = np.zeros(n)
4 x0[7] = 50
5 # show initial objective
6 print('Initial Objective: ' + str(objective(x0)))
7 # bound
8 b1 = (0.0, 1.0)
9 b2 = (-100.0, 200.0)
10 b3 = (35.0, 200.0)
11 bnds = (b1, b1, b1, b2, b2, b2, b2, b3)
12 # constraints
13 con1 = {'type': 'eq', 'fun': constraint1}
14 con2 = {'type': 'eq', 'fun': constraint2}
15 con3 = {'type': 'eq', 'fun': constraint3}
16 con4 = {'type': 'eq', 'fun': constraint4}
17 con5 = {'type': 'eq', 'fun': constraint5}
18 con6 = {'type': 'eq', 'fun': constraint6}
19 cons = ([con1, con2, con3, con4, con5, con6])
20 solution = minimize(objective, x0, bounds = bnds, constraints=cons)
21 print(solution)
```

Optimization Applications

Artificial Intelligent & Machine Learning



Transportation/Distribution Network



Portfolio/Risk Management



Emily Zhu, PhD.

Pricing/Revenue Management



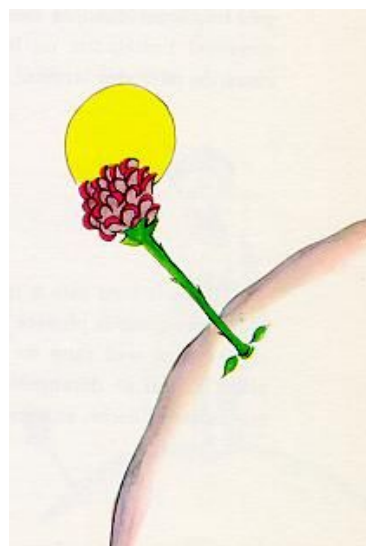
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Before Next Lecture

- **Group Project Guidelines** posted on Canvas
- **Group Project** should move forward



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